

Classical and Quantum Information Geometry via models on W^* -algebras

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Why W^* -algebras?

**Classical
Finite-dimensional**

**Quantum
Finite-dimensional**

**Classical
Infinite-dimensional**

**Quantum
Infinite-dimensional**

First objectives:

- Parametric models
- Information metrics

What is a W^* -algebra?

1. A weakly closed $*$ -subalgebra of $\mathcal{B}(\mathcal{H})$ containing the identity
2. A subalgebra of $\mathcal{B}(\mathcal{H})$ closed under the $*$ -operation and equal to its double commutant
3. A C^* -algebra that has a predual

$1 \Leftrightarrow 2$ Von Neumann double commutant theorem (1930)

$1 \Leftrightarrow 3$ Sakai (1971)

What is a C*-algebra?

$\mathcal{A}$ vector space over $\mathbb{C}$ with a product, a norm $\| \cdot \|$, and an involution * s.t. :

- $(\mathcal{A}, \| \cdot \|)$ Banach space
- $\mathcal{A}$ Banach algebra w.r.t. the norm $\| \cdot \|$ and the product
- $\mathcal{A}$ * -algebra w.r.t. the involution *
- $a \in \mathcal{A}, \quad \|a^*a\| = \|a^*\|\|a\|$

C*-algebras of operators

“Let $\mathcal{A}$ be a C*-algebra, by which we mean a uniformly closed, self-adjoint algebra of bounded operators on a Hilbert space.”

I.-E. Segal(1947): "Irreducible representations of operator algebras", *Bulletin of the American Mathematical Society*, **53** (2): 73–88

Abstract definition

Realization as a
subalgebra of $\mathcal{B}(\mathcal{H})$

The two definitions are equivalent

(Gelfand-Naimark Theorem)

Examples

- The algebra $C_0(\mathbb{R})$ of continuous functions vanishing at infinity is an abelian C^* -algebra, but not a W^* -algebra.
- The algebra $\mathcal{C}(\mathbb{H})$ of compact operators on an Hilbert space is a (non-abelian) C^* -algebra, but not a W^* -algebra.
- The algebra $L^\infty(\Omega, \mu)$ of essentially bounded functions on a measurable space is an abelian W^* -algebra.
- The algebra $\mathcal{B}(\mathbb{H})$ of bounded operators on an Hilbert space is a (non-abelian) W^* -algebra.

Main idea

Let $\mathcal{A}$ be a W^* -algebra

$$\mathcal{A}_+ \hookrightarrow \mathcal{A}_{sa} \hookrightarrow \mathcal{A}$$

$$\mathcal{A}_{sa} = \{a \in \mathcal{A} : a^* = a\}$$

$$\mathcal{A}_+ = \{p \in \mathcal{A} : \exists a \in \mathcal{A} : p = aa^* = a^*a\}$$

$$\mathcal{A}^* \hookleftarrow \mathcal{V} \equiv \mathcal{A}_{sa}^* \hookleftarrow \mathcal{P}$$

$$\mathcal{A}^* = \{\eta : \mathcal{A} \rightarrow \mathbb{C} \text{ cont.}\}$$

$$i : \mathcal{A}_* \hookrightarrow (\mathcal{A}_*)^{**} = \mathcal{A}^*$$

$$\mathcal{P} = \{\omega \in \mathcal{V} : \omega(p) \geq 0 \forall p \in \mathcal{A}_+\}$$

$$\mathcal{P}_N = \mathcal{P} \cap i(\mathcal{A}_*)$$

$$j : \mathcal{M} \rightarrow \mathcal{V} \text{ smooth and s.t. } j(\mathcal{M}) \subseteq \mathcal{P}_N$$

$$(\mathcal{G}(V, W))(m) = \omega_m(\{a_V, a_W\})$$

The Double Tangent Cone

Definition: Let B be a Banach space, let $X \subseteq B$ and $\xi \in X$. An element $\eta \in B$ belongs to the double tangent cone $T_\xi X$ of X at ξ if there exists a positive number ϵ and a C^1 -curve $\gamma : (-\epsilon, \epsilon) \rightarrow B$ s.t. $\gamma(t) \in X$ for all $t \in (-\epsilon, \epsilon)$, $\gamma(0) = \xi$ and $\dot{\gamma}(0) = \eta$

We will see that this is linked to the concept of absolute continuity for functionals.

If X is made of normal functionals, then $T_\xi X$ is made of normal functionals.

Absolutely continuous functionals

Definition: Let $\omega \in \mathcal{P}, \eta \in \mathcal{V}$. Then η is said to be absolutely continuous w.r.t. ω if one of the following, equivalent statements, is true:

- For every $\epsilon > 0$ there exists a $\delta > 0$ s.t. $\omega(a) < \delta \implies |\eta(a)| < \epsilon$ for all $a \in \mathcal{A}_+^1$.
- For every sequence $\{a_n\}_{n \in \mathbb{N}}$ with $a_n \in \mathcal{A}_+^1$, $\lim_n \omega(a_n) = 0 \implies \lim_n \eta(a_n) = 0$.
- For every net $\{a_\lambda\}_{\lambda \in \Lambda}$ with $a_\lambda \in \mathcal{A}_+^1$, $\lim_\lambda \omega(a_\lambda) = 0 \implies \lim_\lambda \eta(a_\lambda) = 0$

Denote the set of all functional that are absolutely continuous w.r.t. ω with AC_ω

A Radon-Nikodym type theorem

Definition: Let $\omega \in \mathcal{P}$. Let us define the set

$$J_\omega = \left\{ \eta \in \mathcal{V} \mid \eta(b) = \frac{1}{2} \omega(ab + ba) \equiv \omega(\{a, b\}) \quad \forall b \in \mathcal{A}_{sa} \right\}$$

For some $a \in \mathcal{A}_{sa}$

ω is normal iff it is continuous in the ultraweak (σ -weak, weak*) topology.

If ω is normal, then $\eta \in \overline{J_\omega}$ is normal too.

Theorem: Let $\omega \in \mathcal{P}$. Holds:

$$\overline{J_\omega} = AC_\omega$$

Why Radon-Nikodym type?

$$\mathcal{A} = L^\infty(\Omega, \mu), \quad \eta \in \mathcal{V}, \quad \eta(b) = \int_\Omega b \, d\mu_\eta$$

$$\eta \in J_\omega \quad \int_\Omega b \, d\mu_\eta = \eta(b) = \frac{1}{2} \omega(ab + ba) = \int_\Omega ab \, d\mu_\omega$$

There exists a μ_ω -integrable function a s.t. $\mu_\eta = a \mu_\omega$

Tangency and absolute continuity (1)

Proposition: *Let $\omega \in \mathcal{P}$. Holds:*

$$AC_\omega \subseteq \overline{T_\omega \mathcal{P}_N}$$

Proof: Set $\omega_t(b) = \omega\left(e^{\frac{ta}{2}} b e^{\frac{ta}{2}}\right)$

$$\omega_0 = \omega \qquad \left. \frac{d\omega_t(b)}{dt} \right|_{t=0} = \frac{1}{2} \omega(ab + ba) \qquad J_\omega \subseteq T_\omega \mathcal{P}$$

$$AC_\omega \subseteq \overline{T_\omega \mathcal{P}_N}$$

It is possible to prove also the inverse relation to conclude

$$\overline{T_\omega \mathcal{P}_N} = AC_\omega = \overline{J_\omega}$$

Parametric models in W^* -algebras

Definition: Let M be a Banach manifold, and $j: M \rightarrow \mathcal{V}$ be a smooth map s.t. $j(M) \subseteq \mathcal{P}_N$ and $T_m j$ is injective for all $m \in M$.

Then the triple $(M, j, \mathcal{A})$ is said to be a parametric model of normal positive linear functionals on $\mathcal{A}$.

$V \in T_m M$, what is $(T_m j)(V)$?

$$\gamma(t) \in M \quad \forall t \in (-\epsilon, \epsilon): \quad \gamma(0) = m, \quad \dot{\gamma}(0) = V \quad (T_m j)(V) = \left. \frac{dj(\gamma(t))}{dt} \right|_{t=0}$$

$$T_m j(T_m M) \subseteq \overline{T_{j(m)} \mathcal{P}_N} = \overline{J_{j(m)}}$$

Generalizing the Fisher-Rao metric

$\eta \in J_\omega$ implies that $\exists a: \eta(c) = \omega(\{a, c\})$

$$G_\omega: J_\omega \times J_\omega \rightarrow \mathbb{R} \qquad G_\omega(\eta_1, \eta_2) = \omega(\{a^1, a^2\})$$

$$G_\omega: \overline{J_\omega} \times J_\omega \rightarrow \mathbb{R} \qquad G_\omega(\eta_1, \eta_2) = \lim_n \omega(\{a_n^1, a^2\})$$

$$G_\omega: J_\omega \times \overline{J_\omega} \rightarrow \mathbb{R} \qquad G_\omega(\eta_1, \eta_2) = \lim_n \omega(\{a^1, a_n^2\})$$

$$G_\omega: \overline{J_\omega} \times \overline{J_\omega} \rightarrow ? \qquad G_\omega(\eta_1, \eta_2) = \lim_n \lim_l \omega(\{a_n^1, a_l^2\})$$

If $T_m j(V) \in J_\omega$ then the model is said to be **regular**.

$$\mathcal{G}: \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathcal{F}(M)$$

$$(\mathcal{G}(V, W))(m) = G_{\omega_m}(T_m j(V_m), T_m j(W_m)) = \omega_m(\{a^{V_m}, a^{W_m}\})$$

If this defines a smooth tensor field on M then the model is said to be **Jordan regular**.

Classical Information Geometry

$$\mathcal{A} = L^\infty(\Omega, \mu, \mathbb{C}), \quad \mathcal{A}_{sa} = L^\infty(\Omega, \mu, \mathbb{R}), \quad \mathcal{A}_* = L^1(\Omega, \mu, \mathbb{C}).$$

$$\mathcal{A}_+ = \{p \in L^\infty(\Omega, \mu, \mathbb{C}) : p \geq 0\}, \quad \mathcal{P} = \{\omega \in \mathcal{A}^* : \omega(p) \geq 0 \ \forall p \in \mathcal{A}_+\}$$

$$\begin{aligned} i: L^1(\Omega, \mu, \mathbb{R}) &\longrightarrow (L^1(\Omega, \mu, \mathbb{R}))^{**} = (L^\infty(\Omega, \mu, \mathbb{R}))^* \\ \text{s.t.} \quad i(L^1(\Omega, \mu, \mathbb{R})) &\subset (L^\infty(\Omega, \mu, \mathbb{R}))^* \end{aligned}$$

$$p: M \longrightarrow (L^\infty(\Omega, \mu, \mathbb{R}))^* : p(M) \subseteq i(L^1(\Omega, \mu, \mathbb{R})) \cap \mathcal{P}$$

Let $(M, p, L^\infty(\Omega, \mu, \mathbb{C}))$ be a regular parametric model of normal positive linear functionals

$$AC_{p(m)} = \{i(f) : f \in L^1(\Omega, p(m)\mu)\}$$

$$p(M) \subseteq i(L^1(\Omega, \mu, \mathbb{R})) \implies AC_{p(m)} = \overline{J_{p(m)}} = \overline{T_{p(m)}\mathcal{P}_N} = K_N^{p(m)}$$

Classical Information Geometry

$$\eta \in J_{p(m)} \Rightarrow \exists a \in L^\infty(\Omega, \mu, \mathbb{R}): \eta(b) = \int_{\Omega} p(m) a b d\mu$$

$$G_{p(m)}(\eta_1, \eta_2) = p(m)(\{a_1, a_2\}) = \int_{\Omega} p(m) a_1 a_2 d\mu$$

$$\{x^j\} \text{ coordinates on } U \subseteq M \text{ open, } \gamma_k(t) = (x_m^1, \dots, x_m^{k-1}, x_m^k + t, x_m^{k+1}, \dots, x_m^n)$$

$$\gamma_k(0) = m, \quad \dot{\gamma}_k(0) = V_k \quad (T_m p(V_k))(b) = \left. \frac{d}{dt} \int_{\Omega} p(\gamma_k(t)) b d\mu \right|_{t=0} = \int_{\Omega} \left. \frac{d}{dt} p(\gamma_k(t)) \right|_{t=0} b d\mu$$

$$\text{if } T_m p(V_k) \in J_{\omega}, \text{ then } \exists a_k \in L^\infty(\Omega, \mu, \mathbb{R}):$$

$$\int_{\Omega} \left. \frac{d}{dt} p(\gamma_k(t)) \right|_{t=0} b d\mu = \int_{\Omega} p(m) a_k b d\mu \quad \forall b \in L^\infty(\Omega, \mu, \mathbb{R})$$

Classical Information Geometry

$$a_k = \frac{1}{p(m)} \frac{d}{dt} p(\gamma_k(t)) \Big|_{t=0} = \frac{1}{p(m)} \frac{\partial p(x)}{\partial x^k} \frac{d\gamma_k(t)}{dt} \Big|_{t=0} = \frac{1}{p(m)} \frac{\partial p(x)}{\partial x^k} \Big|_{x=x_m} V_k$$

$$\mathcal{G}_m(V_j, V_k) = G_{p(m)}(T_m p(V_j), T_m p(V_k)) = p(m)(\{a_j, a_k\})$$

$$= \int_{\Omega} p(m) a_j a_k d\mu = \int_{\Omega} \frac{1}{p(m)} \frac{\partial p(x)}{\partial x^j} \Big|_{x=x_m} \frac{\partial p(x)}{\partial x^k} \Big|_{x=x_m} V_j V_k d\mu$$

$$= \int_{\Omega} p(m) \frac{\partial \log(p(x))}{\partial x^j} \Big|_{x=x_m} \frac{\partial \log(p(x))}{\partial x^k} \Big|_{x=x_m} V_j V_k d\mu$$

$\mathcal{G}_m(V_j, V_k)$ is the Fisher-Rao metric tensor on M

Conclusion and outlooks

- We introduced the statistical manifold and the information metric in the framework of (regular) models on W^* -algebras.
- We recovered the classical case in this more general framework.
- New models, i.e. pure states, unitary models, finite-entropy models, singular models.
- Additional structures, such as the dual connection.

Proof that $K_N^\omega \subseteq AC_\omega$

The unit ball $\mathcal{B}$ in $\mathcal{A}$ is ultraweakly (σ -) compact (Banach-Alaoglu) $\left. \begin{array}{l} \mathcal{A}_+ \text{ is ultraweakly } (\sigma\text{-) closed} \end{array} \right\} \Rightarrow \mathcal{A}_+ \cap \mathcal{B} \text{ is } \sigma\text{-compact}$

Let $C_\epsilon = \{a \in \mathcal{A}_+ \cap \mathcal{B} : |\eta(a)| \geq \epsilon\}$ with $\epsilon \geq 0$ and define $\delta = \inf_{a \in C_\epsilon} \omega(a) \geq 0$

It will suffice to show that $\delta > 0$

η normal (σ -continuous) $\Rightarrow C_\epsilon$ is σ -compact

ω normal (σ -continuous) $\Rightarrow \exists a_0 \in C_\epsilon : \delta = \omega(a_0)$

If $\delta = 0$, then $\eta \in K_\omega$ would imply $\eta(a_0) = 0$, but we have $|\eta(a)| \geq \epsilon$ for all $a \in C_\epsilon$.

Support projections and orthogonal decompositions

Proposition: Let $\mathcal{A}$ be a C^* -algebra and $a \in \mathcal{A}$. Then if $a \in \mathcal{A}_{sa}$, then there is a unique decomposition $a = a_+ - a_-$, where $a_+, a_- \in \mathcal{A}_+$ and $a_+ a_- = 0$.

Proposition: Let $\mathcal{A}$ be a C^* -algebra and $a, b \in \mathcal{A}$. Then if a is well-supported there exists a projection, called the *right (left) support projection* p (q), s.t. $ap = a$ ($qa = a$) and $pb = 0$ ($bq = 0$) whenever $ab = 0$.

If a is normal, i.e. $aa^* = a^*a$, then its left and right support projections coincide.

$T_\omega \mathcal{P}$ is complemented in $\mathcal{V}$

ω normal state, p_ω support projection of ω :

$$\omega(a) = \omega(p_\omega a p_\omega)$$

Decomposition of the W^* -algebra $\mathcal{A}$ in the direct sum: $\mathcal{A} = \mathcal{A}_\omega^{pp} \oplus \mathcal{A}_\omega^{qp} \oplus \mathcal{A}_\omega^{pq} \oplus \mathcal{A}_\omega^{qq}$

Where: $\mathcal{A}_\omega^{pp} = p_\omega \mathcal{A} p_\omega$, $\mathcal{A}_\omega^{pq} = p_\omega \mathcal{A} q_\omega$, $\mathcal{A}_\omega^{qp} = q_\omega \mathcal{A} p_\omega$, $\mathcal{A}_\omega^{qq} = q_\omega \mathcal{A} q_\omega$

$$\overline{T_\omega \mathcal{P}} = K_\omega = \text{Ann} \left((\mathcal{A}_\omega^{pp})^c \right) = (\mathcal{A}_\omega^{pp})^*$$

$$\mathcal{V} \ni \xi = \eta + \eta^c \quad \text{with} \quad \eta \in K_\omega = \overline{T_\omega \mathcal{P}}.$$

Is G_ω well-defined?

Proposition: Let η be a functional in K_ω . Then, there are sequences $\{a_n\}_{n \in \mathbb{N}}$ and $\{a_n\}_{n \in \mathbb{N}}$ s.t.

$$\eta(c) = \lim_n \omega(\{a_n^1, c\})$$

$$\eta(c) = \lim_n \omega(\{a_n^2, c\})$$

for all $c \in \mathcal{A}$ if and only if there is a sequence $\{b_n\}_{n \in \mathbb{N}}$ and an $N \in \mathbb{N}$ s.t.

$$a_n^2 = a_n^1 + b_n$$

for all $n \in \mathbb{N}$.

Is the normality condition needed?

$$\mathcal{A} = C[0,1] \quad \mathcal{P} \ni \omega: C[0,1] \ni f \mapsto \int_0^1 f(t)dt$$

If $f \geq 0$,
then $\omega(f) = 0 \implies f = 0$

Thus every $\eta \in \mathcal{V}$ is in K_ω

$AC_\omega = K_\omega$ would imply that every regular complex Borel measure
on $[0,1]$ is absolutely continuous w.r.t. Lebesgue measure

A Radon-Nikodym type theorem

ω is normal iff it is continuous in the ultraweak (σ -weak, weak*) topology.

If ω is normal, then $\eta \in \overline{J_\omega}$ is normal too.

Theorem: *Let $\omega \in \mathcal{P}$. Holds:*

$$\overline{J_\omega} = AC_\omega$$

Proof: First prove that $\overline{J_\omega} \subseteq AC_\omega$

$$|\eta(b)| = \left| \frac{1}{2} \omega(ab + ba) \right| \leq |\omega(ab)| = |\omega(a\sqrt{b}\sqrt{b})| \leq \sqrt{\omega(aba)\omega(b)} \leq \|\omega\| \|a\| \|b\|^{\frac{1}{2}} \sqrt{\omega(b)}$$

A Radon-Nikodym type theorem

Now to prove that $AC_\omega \subseteq \overline{J_\omega}$, suppose $\eta \in AC_\omega$ and $\eta \notin \overline{J_\omega}$

$$\exists B \in \mathcal{A}_{sa}^{**}: B(\eta) = 1; B(\xi) = 0; \forall \xi \in J_\omega$$

$$\exists a \in \mathcal{A}_{sa}: \xi(b) = \frac{1}{2} \omega(ab + ba); \quad B(\xi) = \hat{\xi}(B) = \frac{1}{2} \hat{\omega}(\hat{a}B + B\hat{a})$$

$$\Omega_B \in \mathcal{A}_{sa}^{***}: \Omega_B(A) = \frac{1}{2} \hat{\omega}(AB + BA); \quad \Omega_B(\hat{a}) = \frac{1}{2} \hat{\omega}(\hat{a}B + B\hat{a}) = B(\xi) = 0;$$

$$\Omega_B(\hat{a}) = 0 \quad \forall \hat{a} \in i(\mathcal{A}_{sa}) \subseteq \mathcal{A}_{sa}^{**}$$

Ω_B is ultraweakly continuous, $i(\mathcal{A}_{sa})$ is σ -dense in $\mathcal{A}_{sa}^{**}$, thus Ω_B is zero on the whole $\mathcal{A}_{sa}^{**}$

A Radon-Nikodym type theorem

Let B_+ and B_- be the orthogonal decomposition of B : $B = B_+ - B_-$, $B_+B_- = 0$.

Let P and Q be the support projections of B_+ and B_- : $PB_+ = B_+P = B_+$, $PB_- = B_-P = 0$

$$\begin{aligned} 0 &= \Omega_B(P) = \frac{1}{2} \widehat{\omega}(BP + PB) = \widehat{\omega}(B_+) \\ 0 &= \Omega_B(Q) = \frac{1}{2} \widehat{\omega}(BQ + QB) = -\widehat{\omega}(B_-) \end{aligned}$$

$$\eta \in AC_\omega \text{ iff } \hat{\eta} \in AC_{\widehat{\omega}} \quad \hat{\eta}(B_+) = \hat{\eta}(B_-) = \hat{\eta}(B) = 0$$

In contradiction with $B(\eta) = 1$

$$\overline{J_\omega} = AC_\omega$$